

Challenge 3: Audio Forensics

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1. Executive Summary

This report describes the recovery of speech from an audio file, `war-intercept.wav`, that had been deliberately jammed. The recording is 4 minutes and 39 seconds long, sampled at 16 kHz in mono. Two continuous wave (CW) tones, at 700 Hz and 2600 Hz, together with harmonics at 1400 Hz and 2800 Hz, were mixed into the audio at a level that made the underlying voice unintelligible.

The jamming was removed with a four-stage Python pipeline. The tones were first deleted directly from the short-time Fourier transform (STFT) of the signal, which avoids the ringing that sharp notch filters introduce. Residual tape hiss was then reduced, the signal was restricted to the speech band, and the voice was amplified with a soft-clipping curve. The cleaned file was transcribed with OpenAI Whisper (small model).

The recovered content is a personal letter dated 30 December 1942, written by a soldier named Charlie to his family. In it he describes being selected, together with Lieutenant Sam Brush, as one of two American officers invited to Christmas dinner with Queen Mary.

2. Tools Used

- **Python 3.11 with a virtual environment:** the language for all scripts, with dependencies isolated for reproducibility.
- **NumPy and the wave module:** array handling, FFT computation, and reading and writing WAV files without ffmpeg.
- **SciPy (scipy.signal):** STFT and inverse STFT for spectral deletion, and Butterworth bandpass filter design.
- **noisereduce:** stationary spectral noise reduction to remove tape hiss.
- **Matplotlib:** FFT spectrum charts, spectrograms and 3D waterfall plots.
- **OpenAI Whisper (small model):** speech-to-text transcription of the cleaned audio.
- **VS Code and PowerShell:** code editing and script execution.

3. Methodology

3.1 Analysis

The raw file was first auditioned, and two loud, steady tones were immediately audible over everything else. To characterise them, a full-band Fast Fourier Transform (FFT) was computed

over the entire recording, converting the waveform from the time domain into a spectrum of frequency against magnitude.

The spectrum was dominated by two very narrow peaks at 700 Hz and 2600 Hz, with smaller peaks at 1400 Hz, 2800 Hz and 4200 Hz. The peaks were exact integer multiples of one another, which is characteristic of synthetic interference and not of environmental noise.

Before settling on a method, the following alternative explanations were tested and rejected:

- Morse code: the switching pattern of the tones carried no encoded message.
- AM, SSB or FM modulation: demodulation at 700 Hz and 2600 Hz produced no speech.
- Speed manipulation: playback at 0.5x and 2.0x revealed nothing.
- Audio reversal: reversed playback contained no speech.
- DTMF signalling: the frequencies do not form any standard DTMF pair.

3.2 Finding

The file was confirmed to contain a human voice recording that had been intentionally masked by two CW tones. Both frequencies have recognisable origins: 700 Hz is a common test frequency in broadcast engineering, and 2600 Hz is the tone that telephone phreakers of the 1960s used to control analogue telephone systems. This suggests the jamming was applied by someone with technical knowledge.

The principal technical finding concerned the removal method. Removing the tones with narrow IIR notch filters leaves a lingering metallic resonance, because filters of that sharpness ring after excitation, and this artefact replaced the original whistle. The solution was to abandon time-domain filtering and to delete the tones in the frequency domain using STFT spectral deletion.

The resulting pipeline consisted of four stages:

- **Stage A, STFT spectral deletion:** the bands at 690-710 Hz, 1390-1410 Hz, 2590-2610 Hz, 2790-2810 Hz and 4190-4210 Hz were reduced to 0.1% of their amplitude, and the audio was rebuilt with the inverse STFT.
- **Stage B, noise reduction:** noisereduce estimated the profile of the background hiss and subtracted it.
- **Stage C, speech bandpass:** a Butterworth filter retained 300-3400 Hz.
- **Stage D, voice boost:** a gain of 15x was applied through a tanh curve, raising quiet passages while softly limiting loud ones.

3.3 Outcome

The jamming tones were completely removed with no audible ringing, and the recovered audio was intelligible and could be transcribed by Whisper over its full duration. The speaker is a soldier named Charlie, writing home on 30 December 1942. He describes how he and Lieutenant Sam Brush were chosen as the two American officers to dine with Queen Mary on Christmas Day, how gifts were exchanged, and how the evening ended feeling like one spent with old friends.

As Whisper is a statistical model, some phrases in the later part of the transcript are imperfect. The transcript is saved in 03transcript.txt.

4. Step by Step Process with Screenshots

Step 1: Environment setup

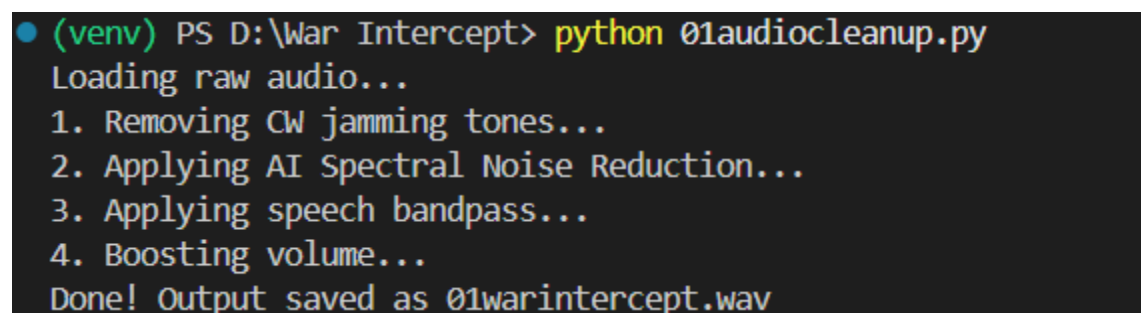
A project folder, D:\War Intercept, was created containing only the original recording. A Python virtual environment was created in it and the required libraries were installed, using the following commands in PowerShell:

```
python -m venv venv
.\venv\Scripts\Activate.ps1
pip install numpy scipy matplotlib noisereduce openai-whisper
python-docx
```

Step 2: Audio cleanup

01audiocleanup.py was written to run the four pipeline stages described above in sequence. Because the tones are erased from the spectrogram and not filtered, no resonance is produced.

```
python 01audiocleanup.py
```



```
(venv) PS D:\War Intercept> python 01audiocleanup.py
Loading raw audio...
1. Removing CW jamming tones...
2. Applying AI Spectral Noise Reduction...
3. Applying speech bandpass...
4. Boosting volume...
Done! Output saved as 01warintercept.wav
```

The output file is 01warintercept.wav.

Step 3: Forensic visualisation

02forensics.py compares the raw and cleaned audio in two charts.

```
python 02forensics.py
```

```
• (venv) PS D:\War Intercept> python 02forensics.py
Generating FFT Spectrum...
Generating Spectrograms...
Done! Outputs saved as 02fig1_fft.png and 02fig2_spectrogram.png
```

Figure 1 shows the FFT spectrum. The sharp peaks at 700 Hz and 2600 Hz in the raw audio are absent from the cleaned audio.

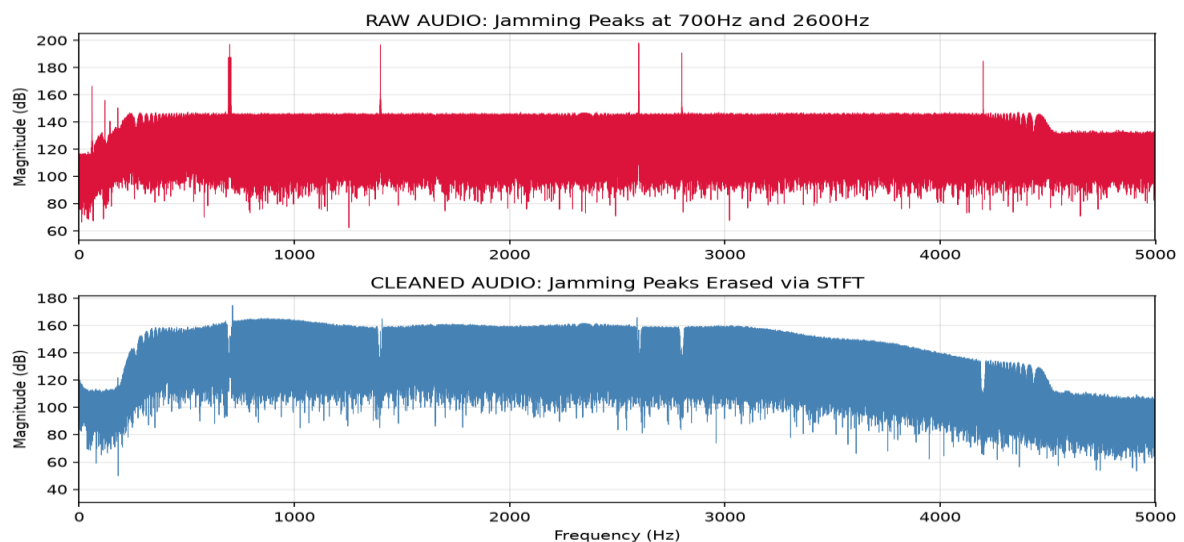


Figure 1: FFT spectrum of raw audio (red) and cleaned audio (blue)

Figure 2 shows the spectrogram. The constant horizontal bands in the raw audio are the jamming tones. In the cleaned audio they are gone, and the irregular pattern of speech is visible.

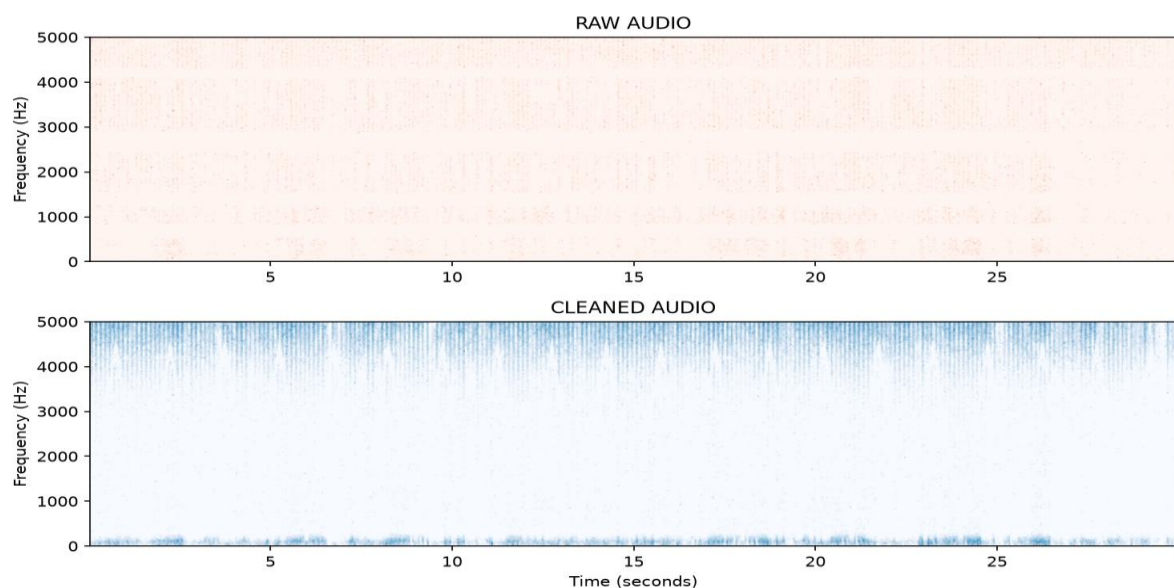


Figure 2: Spectrogram of raw and cleaned audio

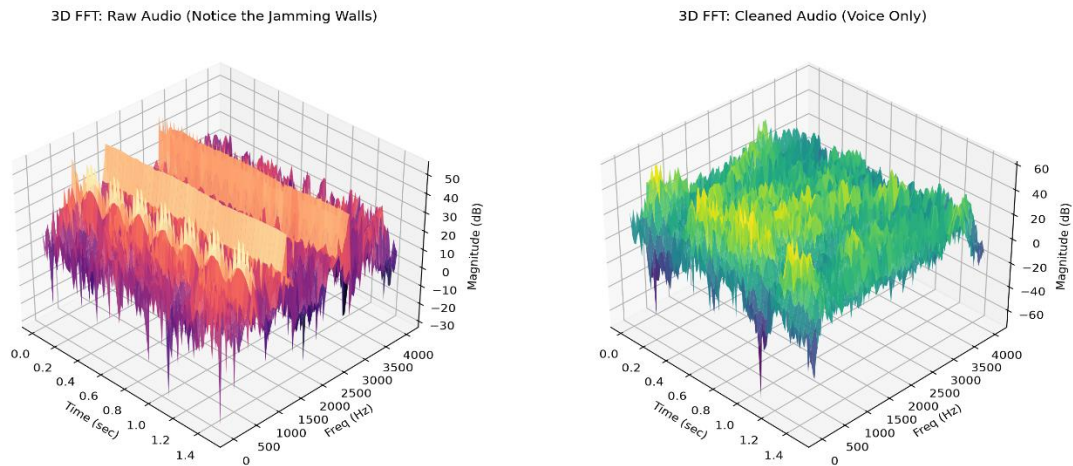


Figure 3: 3-D plotting of Frequency vs Time vs Amplitude

Step 4: Transcription

03transcribe.py passed the cleaned file to Whisper (small model) with `beam_size=10` and `best_of=10`, so that ten candidate transcriptions were generated and the best was kept.

```
python 03transcribe.py
```

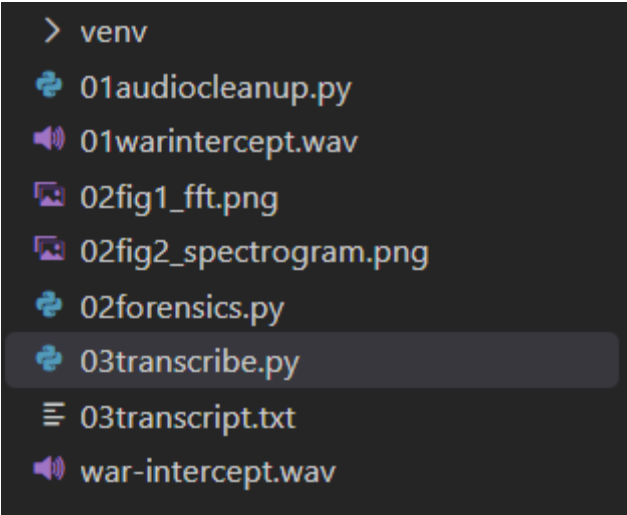
Whisper displayed the warning `FP16 is not supported on CPU; using FP32 instead`. This is not an error. The CPU uses 32-bit instead of 16-bit arithmetic, which does not affect accuracy and only slightly lengthens processing time.

```
(venv) PS D:\War Intercept> $env:PYTHONUTF8="1"; python 03transcribe.py
Loading whisper model (small)...
Transcribing 01warintercept.wav... (This takes a couple of minutes)
D:\War Intercept\venv\Lib\site-packages\whisper\transcribe.py:132: UserWarning: FP16 is not supported on CPU; using FP32 instead
  warnings.warn("FP16 is not supported on CPU; using FP32 instead")
Done! Output saved as 03transcript.txt
```

The output file is `03transcript.txt`.

Step 5: Final output files

On completion, the project folder contained the original recording (`war-intercept.wav`), the three scripts (`01audiocleanup.py`, `02forensics.py`, `03transcribe.py`), the cleaned audio (`01warintercept.wav`), the two charts (`02fig1_fft.png`, `02fig2_spectrogram.png`) and the transcript (`03transcript.txt`).



```
> venv
01audiocleanup.py
01warintercept.wav
02fig1_fft.png
02fig2_spectrogram.png
02forensics.py
03transcribe.py
03transcript.txt
war-intercept.wav
```

Recovered Output Text

The following is the transcript produced by Whisper (small model, beam_size=10, best_of=10), reproduced as generated and saved in 03transcript.txt.

In December 30th, 1942, dear mother, dad and gang, so I'm not supposed to tell you where I've been, I think even the hardest of senses would let a guy tell about Christmas Day as I enjoyed it over here. I have to start with Christmas Eve. It was then I opened your various Christmas parties. Some of the fellows, you rat. They've got any wrong helping and grabbing. Hey, lay off, you guys. These are my cookies. I like cookies. Hey, Charlie, your mother's a swell crook. Well, they opened this one, Charlie. It's small, but they hit it. See, will you look at that. This is what I've been hearing for. What is it? Oh, that's a parking net zone. Isn't that swell? My father sent me that. No, I don't care whether I can get new razor blades or not. You better hang on to that, brother, or I'll be swiping it. You can have anything but my whetstone. I'm gonna hang on to this. Come in. I'm gonna have dinner with the queen. Huh? She's been out in the sun too long. That's what's been no good. Lieutenant Sam Brush and I have been chosen as the two American officers to have Christmas dinner with Queen Mary. Wow! It wasn't pretty, Mother and Dad. We really were invited. You know, Mother, you've always told me it was nice to take along some little things for your hostess. So thanks to your Christmas presents, I was able to put together a small package for the Queen. I admit I hesitated some time over what to put in. And it was partly a royal looking parcel. A box-level perfection paper with six strokes. But I somehow thought the Queen wouldn't mind. Well, when we got there, Your Majesty, Captain George Dudley, Lieutenant Sembrek. I will do Captain Sembrek. Captain Sembrek, thank you. It was very nice of you to come. Nice of us. Darling. I think Christmas is a time when we all want to meet our families. But if you come, the next best thing is to give it to someone else. My family

goes not by the road, too. Oh, yes. Yes, I guess that's wrong. Now, just being careful. Will you accept a little gift from me? Captain, I have one to give and one to you. Thank you. Thank you, Your Majesty. Yes, I do. I do. You have a kind of gift. But that's why I need time to express it to you. Yes, it is, you might have seen. When you laugh, I guess. It isn't really paper clips, but that's the only box I have. Oh, it's a box, it's a box. Let all of them out, please. It's a gift, I shall be. It's a cigarette. It's a deal, sir. I've got a deal with ten knives and a pencil. Oh, well, I hang on to these. It's a question. The shop in Merrick is right the way today. I thought that maybe that's the truth. Maybe the game had a reason. I think it's the very thing you need. You're the thing putting your stuff in a box with some foam. I think it's the very thing I want you to share with me. I'll be there. Join the other guest. I'll be there. Pat him that way. Walk with it on my leg. I would like to hear about the myth of your family. All of it is true. How? When you give me a brush and I left with you tonight, is it worth spending evening with old friends? Here we go. Must close now. Hope you're all well. Well done, Charlie.